WEEK 3 PYTHON & DATA ANALYSIS PROJECT

Sales Data Analysis & Business Insights System

Pandas • Data Cleaning • EDA • Visualization • Insights • OOP Pipeline • GitHub

Overview

You have been provided with a sales dataset containing approximately 5,000 records. Your task is to work as a data analyst: inspect and clean the dataset, analyze it using Pandas, identify meaningful patterns, create visualizations, communicate your findings in a concise professional report, and finally convert the reusable parts of your work into an Object-Oriented data analysis pipeline.

The project should demonstrate not only that you can write Pandas code, but that you can use data to answer meaningful questions and communicate what the results mean.

Project Scenario

A company has provided you with its historical sales data and wants you to investigate business performance. Management is not interested in charts alone. They want to know what is happening in the data, what patterns or problems you can identify, and what reasonable conclusions or recommendations can be made from the evidence.

PHASE 1 — UNDERSTANDING THE DATASET

Before cleaning or analyzing the dataset, first understand what you are working with.

Requirements

1. Import the necessary Python libraries for the analysis.
2. Load the supplied sales dataset into a Pandas DataFrame.
3. Inspect the dataset using appropriate Pandas operations. Your investigation should include the number of rows and columns, column names, data types, sample records, descriptive statistics, missing values, and duplicate records.
4. Use Markdown cells in your Jupyter Notebook to briefly explain what you discovered about the structure and quality of the dataset.

PHASE 2 — DATA CLEANING & PREPARATION

Do not apply cleaning functions blindly. Identify actual problems in the dataset and make reasonable decisions about how they should be handled.

Requirements

5. Investigate and appropriately handle missing values, duplicate records, incorrect data types, inconsistent categorical values, invalid values, date columns, and numerical columns stored incorrectly.

6. Explain significant cleaning decisions using Markdown cells. For example, explain why a column was converted to a different data type or why certain missing values were handled in a particular way.
7. Where appropriate, create useful derived columns that improve the analysis. Examples may include month, year, total sales or revenue, profit margin, order value, or other variables that can reasonably be derived from the supplied dataset.
8. Do not invent information that cannot reasonably be obtained or derived from the dataset.

PHASE 3 — EXPLORATORY DATA ANALYSIS WITH PANDAS

This should be one of the largest technical sections of the project. Use Pandas to investigate the dataset and answer meaningful business questions.

Requirements

9. Use appropriate Pandas techniques such as filtering, sorting, groupby(), aggregation, value_counts(), pivot tables, descriptive statistics, date-based analysis, and comparisons between categories.
10. Answer at least eight meaningful analytical questions using the dataset.
11. Your analysis should investigate relevant areas supported by the dataset, such as overall sales performance, product performance, category performance, regional performance, sales trends over time, customer or customer-segment performance, highest and lowest performing areas, revenue or profitability where available, and unusual patterns or outliers.
12. For each important question, show the code or calculation used to arrive at the answer and explain the result in Markdown.

PHASE 4 — DATA VISUALIZATION

Create visualizations that communicate important findings. A chart should answer a question or make an insight easier to understand.

Requirements

13. Create at least five meaningful charts that explain different findings or insights from your analysis.
14. Choose chart types that suit the question being answered. Possible examples include sales over time, revenue by region, top-performing products, category comparison, customer-segment performance, profit comparison, or monthly sales trends.
15. Every important chart should have an appropriate title, axis labels, readable category labels, and a legend where necessary.
16. Under each major visualization in the notebook, write a short Markdown interpretation explaining what the chart shows and why the finding matters.
17. Save the most important charts as image files so they can be used in your final report.

PHASE 5 — INSIGHTS & BUSINESS INTERPRETATION

The purpose of data analysis is not simply to produce calculations and charts. You must explain what the evidence means.

Requirements

18. Identify at least five substantial, evidence-based insights from the dataset.
19. Support each major insight with calculations, tables, or visualizations.
20. Where appropriate, explain the possible business implication of the finding.
21. Clearly distinguish between what the data directly shows and what is only a possible explanation or interpretation.

Weak example: "The West region has the highest bar."

Stronger example: "The West region generated the highest revenue in the dataset and accounted for a substantial share of total sales. Further analysis should determine whether this performance is driven mainly by transaction volume, average order value, or product mix."

The example above demonstrates the expected style of reasoning. Use the actual numbers and evidence from your own dataset.

PHASE 6 — SALES ANALYSIS REPORT

Produce a concise professional report that communicates the most important parts of your analysis to a non-technical reader.

Report Requirements

22. Export the final report as a PDF.
23. The report must not exceed five pages.
24. Include the project objective and a brief overview of the dataset.
25. Briefly summarize important data-cleaning work.
26. Present the strongest findings using charts and concise written interpretation.
27. Include at least some short, relevant code snippets to demonstrate important parts of your cleaning or analysis process. Do not fill the report with code.
28. Conclude with the major insights, reasonable recommendations, and an overall conclusion.

Suggested Five-Page Report Structure

Section	Suggested Content
Page 1	Project objective, dataset overview, and important cleaning performed.
Pages 2–4	Main analysis and findings. Use the strongest charts with concise interpretation and a few relevant code snippets.
Page 5	Key insights, recommendations, and conclusion.

The notebook contains the detailed technical work. The PDF report should communicate the most important results clearly and concisely.

PHASE 7 — GIT & GITHUB

Requirements

29. Create a Git repository for the project and push it to GitHub.

30. Use multiple meaningful commits during development instead of completing the entire project and making only one final commit.
31. Create a professional README.md explaining the project title, objective, dataset, technologies used, analysis performed, major findings, project structure, and how to run or review the project.
32. Create a requirements.txt file containing the Python packages required by the project.
33. Include an appropriate .gitignore file.
34. Do not upload passwords, API keys, credentials, unnecessary temporary files, or other sensitive information.

Example Commit Messages

Initialize sales analysis project
Add data inspection and cleaning
Add product and regional analysis
Add sales visualizations
Add business insights
Add OOP analysis pipeline
Complete project documentation

PHASE 8 — REQUIRED OOP DATA ANALYSIS PIPELINE

This is the final stage of the project. After completing the exploratory analysis and identifying the cleaning and analysis operations that work, convert the reusable parts of your work into an Object-Oriented data analysis pipeline.

Scenario

Imagine the company gives you another compatible sales CSV file next month. You should not have to manually copy dozens of notebook cells and repeat the entire process. Your goal is to make the important parts of your cleaning and analysis workflow reusable.

Requirements

35. Create a separate Python module for the reusable analysis pipeline, for example sales_analyzer.py.
36. Create an appropriate class representing the analysis system.
37. The class should accept the dataset or the path to the dataset when an object is created.
38. Use appropriate attributes to maintain information needed by the analysis.
39. Convert suitable operations from the notebook into reusable methods.
40. Your pipeline should be capable of performing important operations such as loading data, inspecting data, cleaning data, handling duplicates, handling missing values, correcting data types, performing necessary transformations, generating useful summaries, performing major sales analyses, and saving the cleaned dataset.
41. Import your class into the Jupyter Notebook and demonstrate that the pipeline works.
42. Design the pipeline so that another compatible sales dataset could reasonably be supplied without rewriting the entire analysis.

You are expected to decide how to structure the class and what the methods should be called. The assignment defines what the pipeline should accomplish, but you should think about the design yourself.

Expected Project Files

```
sales-analysis-project/
│
├── data/
│   ├── sales_data.csv
│   └── cleaned_sales_data.csv
│
├── charts/
│   ├── monthly_sales.png
│   ├── regional_sales.png
│   ├── product_performance.png
│   └── ...
│
├── sales_analysis.ipynb
├── sales_analyzer.py
├── sales_analysis_report.pdf
├── README.md
├── requirements.txt
└── .gitignore
```

Your project does not have to reproduce this folder structure character-for-character, but all required project components should be present and clearly organized.

Concepts Your Project Must Demonstrate

- Pandas DataFrames and Series
- Dataset inspection and descriptive statistics
- Data cleaning and missing-value handling
- Duplicate handling and data-type conversion
- Filtering, sorting, grouping, and aggregation
- Pivot tables or equivalent summary techniques where appropriate
- Data transformation and feature creation
- Matplotlib and/or Seaborn visualization
- Analytical reasoning and business insight communication
- Markdown documentation in Jupyter Notebook
- Classes and objects
- Attributes and methods
- Reusable data-processing and analysis code
- Importing your own Python module into a notebook
- Git and GitHub
- README documentation
- Dependency management using requirements.txt

Research Component

You are allowed and expected to research anything you do not already know. This project may require you to independently investigate topics such as:

- Advanced Pandas operations
- Pivot tables
- Date and time handling
- Missing-value strategies
- Matplotlib or Seaborn customization
- Saving charts as image files
- Markdown in Jupyter Notebook
- Creating and exporting a concise PDF report
- Python classes and reusable methods
- Importing your own Python modules
- requirements.txt
- .gitignore
- Git and GitHub workflows

You may consult documentation, search engines, tutorials, or AI tools, but you must understand and be able to explain every part of the work you submit.

What You Should Be Prepared to Explain

- How you inspected and cleaned the dataset.
- Why you made important cleaning decisions.
- How you answered your analytical questions.
- Why you selected particular charts.
- What your strongest insights are and what evidence supports them.
- How your OOP pipeline works.
- Why the pipeline is more reusable than leaving all the code as separate notebook cells.
- How you used Git and GitHub during the project.

Important Notes

- Do not simply copy a complete analysis or solution from the Internet or an AI tool.
- Research is encouraged, but you must understand what you implement.
- Your analysis should be driven by the actual columns and information available in the supplied dataset.
- Do not invent conclusions that the data does not support.
- Test your code and rerun the notebook from beginning to end before finalizing the project.
- Keep the PDF report concise: maximum five pages.
- Use at least five meaningful charts and at least five substantial insights.
- The OOP data analysis pipeline is a required final stage, not an optional bonus.
- Your project should be organized well enough that another person can understand how to review or run it.

Good luck!

Analyze carefully, explain your reasoning, build reusable code, and make sure you understand every part of your project.